Applied Physics -1

Motion and Relative Motion - Lecture 04 (Module -01)

Relative Motion

Definition

Relative motion is the motion of one object as observed from another object that may itself be moving.
The position, velocity, and acceleration of an object depend on the reference frame of the observer.
Example:
A person sitting inside a moving train is at rest relative to the train, but is moving relative to a person standing on the ground.
Therefore, motion is relative to the observer.

Reference Frame

A reference frame is a coordinate system or viewpoint with respect to which the position and motion of an object are measured.
For example, consider two reference frames:

- A stationary observer on the ground.
- An observer inside a moving train.

The same object can have different velocities when observed from these two reference frames.

Relative Position

Suppose the position vectors of objects A and B with respect to a fixed reference frame are $\vec{r}_A$ and $\vec{r}_B$.
The position of A relative to B is

$$\boxed{\vec{r}_{A/B} = \vec{r}_A - \vec{r}_B}$$

Similarly,

$$\boxed{\vec{r}_{B/A} = \vec{r}_B - \vec{r}_A}$$

Therefore,

$$\boxed{\vec{r}_{B/A} = -\vec{r}_{A/B}}$$

Relative Velocity

Relative velocity is the velocity of one object as observed from another object.
If the velocities of objects A and B are $\vec{v}_A$ and $\vec{v}_B$, then the velocity of A relative to B is

$$\boxed{\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B}$$

Similarly,

$$\boxed{\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A}$$

Hence,

$$\boxed{\vec{v}_{B/A} = -\vec{v}_{A/B}}$$

Relative Acceleration

The relative acceleration of A with respect to B is

$$\vec{a}_{A/B} = \frac{d\vec{v}_{A/B}}{dt}$$

Since

$$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B,$$

we obtain

$$\boxed{\vec{a}_{A/B} = \vec{a}_A - \vec{a}_B}$$

Relative Motion in One Dimension

For one-dimensional motion, a sign convention is used.
If two objects move in the same direction,

$$\boxed{v_{\text{rel}} = |v_A - v_B|}$$

If two objects move in opposite directions,

$$\boxed{v_{\text{rel}} = |v_A + v_B|}$$

Thus,

$$\boxed{v_{\text{rel}} = \begin{cases} |v_A - v_B|, & \text{same direction,} \\ |v_A + v_B|, & \text{opposite directions.} \end{cases}}$$

Time of Meeting

Suppose two objects are separated by a distance d and move toward each other with speeds v_A and v_B.
Their relative speed is

$$v_{\text{rel}} = v_A + v_B.$$

Therefore,

$$\boxed{t = \frac{d}{v_A + v_B}}$$

If two objects move in the same direction, with one object chasing the other,

$$v_{\text{rel}} = v_{\text{faster}} - v_{\text{slower}}.$$

Hence,

$$\boxed{t = \frac{d}{v_{\text{faster}} - v_{\text{slower}}}}$$

Example

Two cars are moving in the same direction with velocities

$$v_A = 25\,\text{m}\,\text{s}^{-1}, \qquad v_B = 15\,\text{m}\,\text{s}^{-1}.$$

The velocity of car A relative to car B is

$$v_{A/B} = v_A - v_B.$$

Therefore,

$$v_{A/B} = 25 - 15$$

$$\boxed{v_{A/B} = 10\,\mathrm{m\,s^{-1}}}$$

Thus, an observer in car B sees car A moving with a velocity of $10\,\mathrm{m\,s^{-1}}$.

Projectile Motion

Definition

Projectile motion is the two-dimensional motion of an object projected into the air and moving under the influence of gravity, when air resistance is neglected.
Examples include:

- A ball thrown at an angle.
- A stone projected into the air.
- A football kicked into the air.
- A projectile launched horizontally from a height.

Projectile motion is a combination of two independent motions:

$$\boxed{\text{Horizontal motion} + \text{Vertical motion}}$$

The horizontal motion has constant velocity, while the vertical motion has constant acceleration due to gravity.

Initial Velocity

Suppose a projectile is launched with initial velocity u at an angle θ above the horizontal.
The velocity is resolved into two perpendicular components.
Horizontal component:

$$\boxed{u_x = u\cos\theta}$$

Vertical component:

$$\boxed{u_y = u\sin\theta}$$

Therefore,

$$\boxed{\vec{u} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}}$$

Horizontal Motion

When air resistance is neglected, there is no horizontal acceleration.
Therefore,

$$\boxed{a_x = 0}$$

Hence, the horizontal velocity remains constant:

$$\boxed{v_x = u\cos\theta}$$

The horizontal displacement after time t is

$$x = v_x t.$$

Therefore,

$$\boxed{x = u\cos\theta\,t}$$

Vertical Motion

Taking upward as positive, acceleration due to gravity is

$$\boxed{a_y = -g}$$

The vertical velocity after time t is

$$v_y = u_y + a_y t.$$

Therefore,

$$\boxed{v_y = u\sin\theta - gt}$$

The vertical displacement is

$$y = u_y t + \frac{1}{2}a_y t^2.$$

Hence,

$$\boxed{y = u\sin\theta\,t - \frac{1}{2}gt^2}$$

Equation of Trajectory

The horizontal displacement is

$$x = u\cos\theta\,t.$$

Therefore,

$$t = \frac{x}{u\cos\theta}.$$

Substituting this into

$$y = u\sin\theta\,t - \frac{1}{2}gt^2,$$

we get

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}.$$

Therefore,

$$\boxed{y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}}$$

Since the equation contains x^2, the path of the projectile is a parabola.

Time of Flight

For a projectile that lands at the same level from which it was launched,

$$y = 0.$$

Therefore,

$$0 = u\sin\theta\,t - \frac{1}{2}gt^2.$$

Taking the non-zero solution,

$$\boxed{T = \frac{2u\sin\theta}{g}}$$

where T is the total time of flight.

Maximum Height

At the maximum height, the vertical velocity becomes zero. Therefore,

$$v_y = 0.$$

Using

$$v_y^2 = u_y^2 - 2gH,$$

we get

$$0 = (u\sin\theta)^2 - 2gH.$$

Hence,

$$\boxed{H = \frac{u^2\sin^2\theta}{2g}}$$

Horizontal Range

The horizontal range is the total horizontal distance travelled by the projectile.
It is given by

$$R = u\cos\theta\, T.$$

Substituting

$$T = \frac{2u\sin\theta}{g},$$

we obtain

$$R = \frac{2u^2\sin\theta\cos\theta}{g}.$$

Using

$$2\sin\theta\cos\theta = \sin 2\theta,$$

we obtain

$$\boxed{R = \frac{u^2\sin 2\theta}{g}}$$

Maximum Range

The maximum value of $\sin 2\theta$ is 1.
Therefore,

$$\sin 2\theta = 1.$$

Hence,

$$2\theta = 90^\circ$$

and

$$\boxed{\theta = 45^\circ}$$

The maximum range is therefore

$$\boxed{R_{\max} = \frac{u^2}{g}}$$

Important Projectile Relations

For a projectile launched and landing at the same level,

$$\boxed{T = \frac{2u\sin\theta}{g}}$$

$$\boxed{H = \frac{u^2\sin^2\theta}{2g}}$$

$$\boxed{R = \frac{u^2\sin 2\theta}{g}}$$

$$\boxed{R_{\max} = \frac{u^2}{g} \quad \text{at} \quad \theta = 45^\circ}$$

Breaking Down Multi-D Motion

Definition

Multi-dimensional motion occurs when an object moves simultaneously in more than one direction.
For two-dimensional motion, the motion is resolved into two perpendicular components:

$$\boxed{\text{2-D motion} = \text{x-direction motion} + \text{y-direction motion}}$$

This process is called **component analysis**.

Why Components Are Used

A two-dimensional problem can often be difficult to solve directly.
By resolving the motion into perpendicular components, the problem can be treated as two simpler one-dimensional problems.
Thus,

$$\boxed{\text{Complex 2-D motion} \longrightarrow \text{Simple x-motion} + \text{Simple y-motion}}$$

Resolving a Vector

Consider a vector $\vec{A}$ of magnitude A making an angle θ with the positive x-axis.
The horizontal component is

$$\boxed{A_x = A\cos\theta}$$

The vertical component is

$$\boxed{A_y = A\sin\theta}$$

Therefore,

$$\boxed{\vec{A} = A_x\hat{i} + A_y\hat{j}}$$

or

$$\boxed{\vec{A} = A\cos\theta\,\hat{i} + A\sin\theta\,\hat{j}}$$

Finding Magnitude from Components

If the components A_x and A_y are known, the magnitude of the vector is obtained using the Pythagorean theorem:

$$\boxed{A = \sqrt{A_x^2 + A_y^2}}$$

Finding Direction

The direction of the vector with respect to the positive x-axis is

$$\tan\theta = \frac{A_y}{A_x}.$$

Therefore,

$$\boxed{\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)}$$

The signs of A_x and A_y must be considered to determine the correct quadrant.

Velocity Components

A velocity vector can be resolved into two components. For velocity v making an angle θ with the horizontal,

$$\boxed{v_x = v\cos\theta}$$

and

$$\boxed{v_y = v\sin\theta}$$

Therefore,

$$\boxed{\vec{v} = v_x\hat{i} + v_y\hat{j}}$$

Acceleration Components

Similarly, acceleration can be resolved into components:

$$\boxed{a_x = a\cos\theta}$$

$$\boxed{a_y = a\sin\theta}$$

Therefore,

$$\boxed{\vec{a} = a_x\hat{i} + a_y\hat{j}}$$

For projectile motion,

$$\boxed{a_x = 0}$$

and

$$\boxed{a_y = -g}$$

when upward is chosen as positive.

Position Components

The position vector of a particle in two dimensions is

$$\boxed{\vec{r} = x\hat{i} + y\hat{j}}$$

For constant acceleration,

$$\boxed{\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2}$$

Resolving this equation into components gives

$$\boxed{x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2}$$

and

$$\boxed{y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2}$$

For constant acceleration, the three equations of motion in two dimensions are
For the x-axis:

$$\boxed{v_x = u_x + a_x t}$$

$$\boxed{x = x_0 + u_x t + \frac{1}{2}a_x t^2}$$

$$\boxed{v_x^2 = u_x^2 + 2a_x(x - x_0)}$$

For the y-axis:

$$\boxed{v_y = u_y + a_y t}$$

$$\boxed{y = y_0 + u_y t + \frac{1}{2}a_y t^2}$$

$$\boxed{v_y^2 = u_y^2 + 2a_y(y - y_0)}$$

These equations are used to solve two-dimensional motion problems.

General Method for Solving Multi-D Motion

The following procedure can be used:

1. Draw a clear diagram of the motion.
2. Choose the x-axis and y-axis.
3. Select the positive direction of each axis.
4. Resolve the initial velocity into components.

$$v_{0x} = v_0\cos\theta, \qquad v_{0y} = v_0\sin\theta.$$

5. Resolve the acceleration into x and y components.
6. Solve the motion along the x-direction.
7. Solve the motion along the y-direction.
8. Use the same time t for both directions.
9. Combine the components if the final magnitude or direction is required.

Worksheet Questions

1. Two cars A and B are moving in the same direction with velocities $25\,\mathrm{m\,s^{-1}}$ and $15\,\mathrm{m\,s^{-1}}$, respectively. Determine the velocity of car A relative to car B.

2. Two trains are moving in opposite directions with speeds $20\,\mathrm{m\,s^{-1}}$ and $30\,\mathrm{m\,s^{-1}}$. Find their relative velocity.

3. A particle starts from rest and moves with a constant acceleration of $4\,\mathrm{m\,s^{-2}}$ for $8\,\mathrm{s}$. Determine its final velocity and displacement.

4. A car travelling at $20\,\mathrm{m\,s^{-1}}$ accelerates uniformly at $3\,\mathrm{m\,s^{-2}}$. Find its velocity after $6\,\mathrm{s}$ and the distance travelled during this time.

5. A stone is dropped from a height and reaches the ground after $4\,\mathrm{s}$. Taking $g = 9.8\,\mathrm{m\,s^{-2}}$, determine the height from which it was dropped.

6. A projectile is launched with an initial speed of $20\,\mathrm{m\,s^{-1}}$ at an angle of 30° above the horizontal. Determine its time of flight and horizontal range. Take $g = 9.8\,\mathrm{m\,s^{-2}}$.

7. A projectile is fired with a speed of $40\,\mathrm{m\,s^{-1}}$ at an angle of 45°. Determine its maximum height and horizontal range. Take $g = 9.8\,\mathrm{m\,s^{-2}}$.

8. A particle is projected with an initial velocity of $30\,\mathrm{m\,s^{-1}}$ at an angle of 60° with the horizontal. Determine the horizontal and vertical components of its initial velocity.

9. A velocity vector has components

$$v_x = 12\,\mathrm{m\,s^{-1}}, \qquad v_y = 5\,\mathrm{m\,s^{-1}}.$$

Determine its magnitude and direction with respect to the positive x-axis.

10. A particle has initial velocity components

$$v_{0x} = 10\,\mathrm{m\,s^{-1}}, \qquad v_{0y} = 20\,\mathrm{m\,s^{-1}}.$$

If the acceleration components are

$$a_x = 2\,\mathrm{m\,s^{-2}}, \qquad a_y = -9.8\,\mathrm{m\,s^{-2}},$$

determine its position components after 3 s, assuming it starts from the origin.

11. A ball is projected from the ground with a velocity of $25\,\mathrm{m\,s^{-1}}$ at an angle of 37°. Determine:

(a) Initial horizontal velocity
(b) Initial vertical velocity
(c) Time of flight
(d) Maximum height
(e) Horizontal range

Take

$$\sin 37^\circ = 0.6, \qquad \cos 37^\circ = 0.8, \qquad g = 10\,\mathrm{m\,s^{-2}}.$$

12. Two particles A and B are initially separated by 500 m. Particle A moves with a velocity of $20\,\mathrm{m\,s^{-1}}$ while particle B moves toward A with a velocity of $30\,\mathrm{m\,s^{-1}}$.

Determine:

(a) Relative velocity of A with respect to B
(b) Time taken for them to meet